

# Mathematics

Stage 8

Paper 1

**2025**

Cambridge Lower Secondary Progression Test

## Mark Scheme

**General guidance on marking****Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

**Square brackets in mark scheme**

When square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

| Question | Answer              | Mark | Part marks | Guidance |
|----------|---------------------|------|------------|----------|
| 5        | 19.7 or 19.6[58...] | 1    |            |          |

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

**Accept**

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

**Do not accept**

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

### Number and place value

The table shows various general rules in terms of acceptable decimal answers.

| Accept                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|
| Accept omission of leading zero if answer is clearly shown, e.g.<br><b>.675</b>                                                            |
| Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g.<br><b>0.7000</b> |
| Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. <b>0,638</b>                           |

### Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

|                                                                                                       | Accept                                                                                                                                                                                                             | Do not accept                                            |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| If the unit is given on the answer line, e.g.<br>..... m                                              | Correct conversions, provided the unit is stated unambiguously, e.g. .....185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer) | .....185..... m<br>.....1850..... m etc.                 |
| If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres' | 1.85<br>1 m 85 cm                                                                                                                                                                                                  | 185; 1850<br>Any conversions to other units, e.g. 185 cm |

**Money**

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

|                                                                                         | Accept                                                                                                                | Do not accept                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If the amount is in dollars and cents, the answer should be given to two decimal places | \$0.30<br><br>For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00    | \$0.3<br><br>\$09 or \$09.00                                                                                                                                                                                     |
| If units are not given on the answer line                                               | Any unambiguous indication of the correct amount, e.g.<br>30 cents; 30 c<br>\$0.30; \$0–30; \$0=30; \$00:30           | 30 or 0.30 without a unit<br><br>\$30; 0.30 cents<br><br>Ambiguous answers, e.g.<br>\$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number) |
| If \$ is shown on the answer line                                                       | All unambiguous indications, e.g.<br>\$.....0.30.....;<br>\$.....0-30.....;<br>\$.....0=30.....;<br>\$.....00:30..... | \$.....30.....<br><br>Ambiguous answers, e.g.<br>\$.....30 cents.....;<br>\$.....0.30 cents.....<br><b>unless</b> units on the answer line have been deleted, e.g.<br>\$.....30 cents.....                       |
| If cents is shown on the answer line                                                    | .....30.....cents                                                                                                     | .....0.30.....cents<br><br>Ambiguous answers, e.g.<br>.....\$30 .....cents;<br>.....\$0.30 .....cents<br><b>unless</b> units on the answer line have been deleted, e.g.<br>.....\$0.30.....cents                 |

**Duration**

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

| Accept                                                                                                                                                                                  | Do not accept                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g.<br>2 hours 30 minutes; 2 h 30 m; 02 h 30 m | Incorrect or ambiguous formats, e.g.<br>2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3;<br>2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes) |
| Any correct conversion with appropriate units, e.g.<br>2.5 hours; 150 mins<br><b>unless</b> the question specifically asks for time given in hours and minutes                          | 02:30 (as this is a 24-hour clock time, not a time interval)<br><br>2.5; 150                                                                                                     |

**Time**

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

|                                             | Accept                                                                                                                                                                                                                                                                                      | Do not accept                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| If the answer is required in 24-hour format | Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g.<br>07:30 with any separator in place of the colon, e.g.<br>07 30; 07,30; 07-30; 0730                                                                                                       | 7:30<br>7:30 am<br>7 h 30 m<br>7:3<br>730<br>7.30 pm<br>073<br>07.3 |
| If the answer is required in 12-hour format | Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g.<br>7:30 am with any separator in place of the colon, e.g.<br>7 30 am; 7.30 am; 7-30 am<br><br>7.30 in the morning<br><br>Half past seven (o'clock) in the morning<br><br>Accept am or a.m. | Absence of am or pm<br>1930 am<br>7 h 30 m<br>7:3<br>730<br>7.30 pm |

**Algebra**

The table shows acceptable and unacceptable versions of the answer  $3x - 2$

| Accept                                              | Do not accept                                      |
|-----------------------------------------------------|----------------------------------------------------|
| $x^3 - 2$ ; $3 \times x - 2$                        | $3x + -2$ if it is supposed to be in simplest form |
| Case change in letters                              |                                                    |
| Changes in letters as long as there is no ambiguity |                                                    |

Accept extra brackets when factorising, e.g.  $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

**Inequalities**

The table shows acceptable and unacceptable versions of various answers.

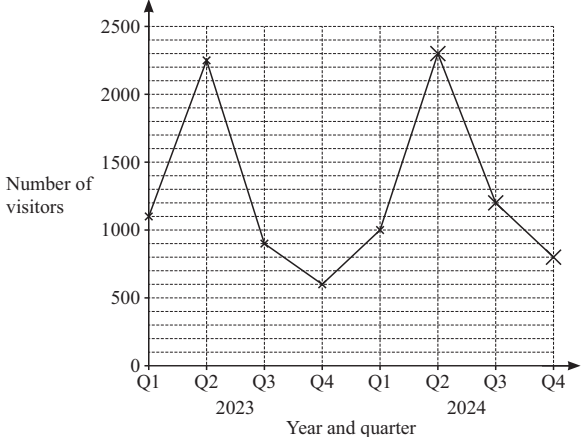
| For the following  | Accept                   | Do not accept                                                               |
|--------------------|--------------------------|-----------------------------------------------------------------------------|
| For $6 \leq x < 8$ | $[6, 8)$                 | $< x <$                                                                     |
| For $x \leq -2$    | $(-\infty, -2]$          | $x < -2$                                                                    |
| For $x > 3$        | $(3, \infty)$<br>$3 < x$ | Just '3' written on the answer line, even if $x > 3$ appears in the working |

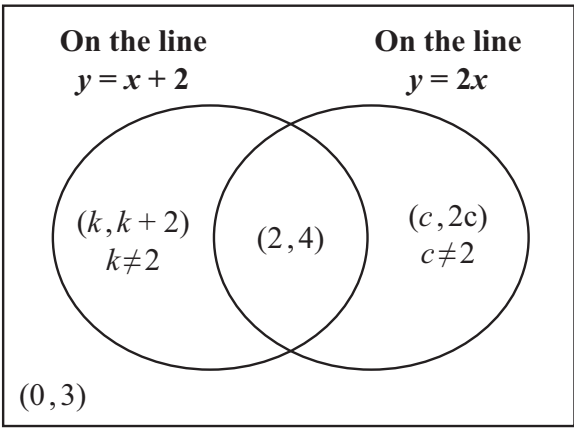
**Plotting points**

The table shows acceptable and unacceptable ways to plot points.

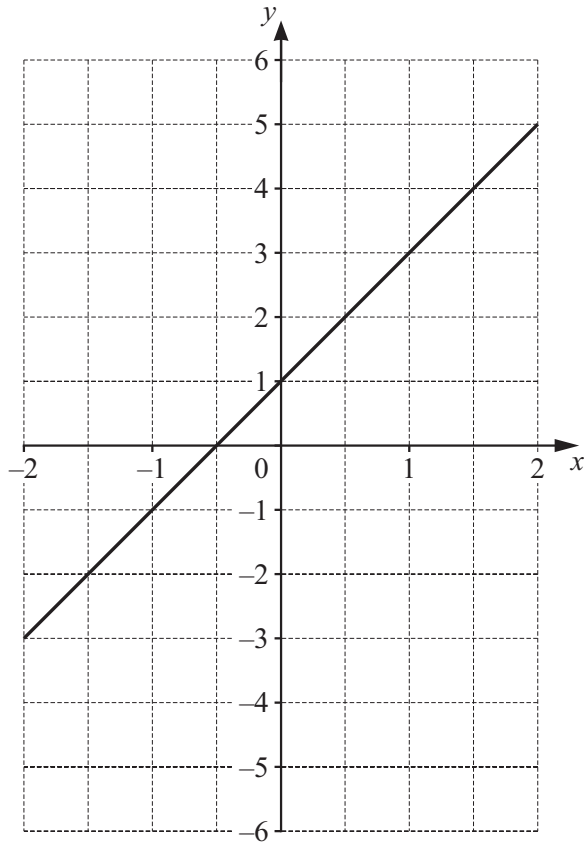
| Accept                                                                                 | Do not accept                                                                   |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer          | A horizontal line and vertical line from the axes meeting at the required point |
| The graph line passing through a point implies the point even though there is no cross |                                                                                 |

| Question        | Answer                                                                                                                                                                                                                                                                                   | Marks      | Part Marks                                      | Guidance                                                |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------|---------------------------------------------------------|----------------|------|----|----------------|------|------|-----------------|-----|-----|---|--------------------------------------------------------|------------------------------|
| 1               | $3\frac{1}{6}$ correct answer only                                                                                                                                                                                                                                                       | 2          | Award 1 mark for $3\frac{2}{12}$ or equivalent. | Or equivalent, e.g. $\frac{38}{12}$ , $\frac{19}{6}$    |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| 2               | $\frac{0.72}{72}$                                                                                                                                                                                                                                                                        | 2          | Award 1 mark for each correct answer.           | Answers must be in the correct order.                   |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| 3               | <table><tr><th>Fraction</th><th>Decimal</th><th>Percentage</th></tr><tr><td><math>\frac{3}{10}</math></td><td>0.03</td><td>3%</td></tr><tr><td><math>\frac{9}{20}</math></td><td>0.45</td><td>4.5%</td></tr><tr><td><math>\frac{75}{50}</math></td><td>1.5</td><td>15%</td></tr></table> | Fraction   | Decimal                                         | Percentage                                              | $\frac{3}{10}$ | 0.03 | 3% | $\frac{9}{20}$ | 0.45 | 4.5% | $\frac{75}{50}$ | 1.5 | 15% | 2 | Award 1 mark for <b>two</b> out of three correct rows. | Accept any clear indication. |
| Fraction        | Decimal                                                                                                                                                                                                                                                                                  | Percentage |                                                 |                                                         |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| $\frac{3}{10}$  | 0.03                                                                                                                                                                                                                                                                                     | 3%         |                                                 |                                                         |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| $\frac{9}{20}$  | 0.45                                                                                                                                                                                                                                                                                     | 4.5%       |                                                 |                                                         |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| $\frac{75}{50}$ | 1.5                                                                                                                                                                                                                                                                                      | 15%        |                                                 |                                                         |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| 4               | + 4                                                                                                                                                                                                                                                                                      | 1          |                                                 | Accept correct equivalents, e.g. plus 4, add four, etc. |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| 5               | $6\frac{6}{7}$ or $\frac{48}{7}$                                                                                                                                                                                                                                                         | 1          |                                                 |                                                         |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| 6               | $\frac{6}{6}$                                                                                                                                                                                                                                                                            | 2          | Award 1 mark for each correct answer.           |                                                         |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |
| 7               | $\frac{1}{5}$                                                                                                                                                                                                                                                                            | 1          |                                                 | Accept any clear indication.                            |                |      |    |                |      |      |                 |     |     |   |                                                        |                              |

| Question | Answer                                                                                                                                                             | Marks | Part Marks                                                                                                | Guidance                                |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 8(a)     | <p>The three remaining points correctly plotted and joined with ruled lines.</p>  | 2     | Award 1 mark for correctly plotting at least <b>two</b> points from (Q2, 2300), (Q3, 1200) and (Q4, 800). | Mark intention.                         |
| 8(b)     | 1100                                                                                                                                                               | 1     |                                                                                                           |                                         |
| 9(a)     | <p>A correct explanation, e.g.<br/>There are 5 green counters out of a total of 15 counters and <math>\frac{5}{15} = \frac{1}{3}</math></p>                        | 1     |                                                                                                           | Accept equivalent correct explanations. |
| 9(b)     | <p>A correct explanation, e.g.<br/>He is equally likely to take a green counter from bag B [as the probability is still <math>\frac{1}{3}</math>].</p>             | 1     |                                                                                                           | Accept equivalent correct explanations. |
| 10       | -5<br>12                                                                                                                                                           | 2     | Award 1 mark for each correct answer.                                                                     | Answers must be in the correct order.   |

| Question | Answer                                                                             | Marks | Part Marks                                                                                                                                                          | Guidance                                               |
|----------|------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 11(a)    | Correct bearing of $070^\circ$ from point $X$ drawn.                               | 1     |                                                                                                                                                                     | Tolerance $\pm 2^\circ$ .                              |
| 11(b)    | $240(^\circ)$                                                                      | 1     |                                                                                                                                                                     |                                                        |
| 12(a)    | B                                                                                  | 1     |                                                                                                                                                                     |                                                        |
| 12(b)    | 8                                                                                  | 1     |                                                                                                                                                                     |                                                        |
| 13       | 10                                                                                 | 1     |                                                                                                                                                                     |                                                        |
| 14       | $13(\text{cm}^2)$                                                                  | 3     | <p>Award 1 mark for <math>\frac{1}{2} \times 5 \times (6 + 16)</math> or better</p> <p><b>and</b></p> <p>Award 1 mark for <math>3 \times 14</math> <b>or</b> 42</p> | Or better, e.g. $\frac{1}{2} \times 5 \times 22$ or 55 |
| 15       |  | 3     | Award 1 mark for each correct numerical coordinate pair.                                                                                                            |                                                        |
| 16       | 32                                                                                 | 2     | Award 1 mark for $[-]\sqrt{144}$ or better.                                                                                                                         | Or better, e.g. $[-]12$ , $[-]18$                      |

| Question | Answer        | Marks | Part Marks                                                                                                                                                                                                                                                                     | Guidance               |
|----------|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 17       | $9x + 5$ (cm) | 2     | Award 1 mark for either of these: <ul style="list-style-type: none"><li>• a correct unsimplified expression</li><li>• <math>3x</math> <b>and</b> <math>x + 5</math> seen</li></ul>                                                                                             | e.g. $2x + 6x + x + 5$ |
| 18       | 10 (cm)       | 2     | Award 1 mark for any one of these: <ul style="list-style-type: none"><li>• <math>12 \text{ cm}^2</math> for area of bottom right rectangle</li><li>• 4 cm or 40 mm for length of bottom right rectangle</li><li>• 6 cm or 60 mm for length of bottom left rectangle.</li></ul> |                        |

| Question | Answer                                                                                                                                                                   | Marks | Part Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Guidance |    |   |   |     |    |   |   |                                                                               |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----|---|---|-----|----|---|---|-------------------------------------------------------------------------------|
| 19       | <p>Correct ruled line drawn between <math>(-2, -3)</math> and <math>(2, 5)</math></p>  | 3     | <p>Award 2 marks for any one of these:</p> <ul style="list-style-type: none"><li>a correctly completed table</li></ul> <table border="1" data-bbox="1068 346 1435 426"><tr><td><math>x</math></td><td>-2</td><td>0</td><td>2</td></tr><tr><td><math>y</math></td><td>-3</td><td>1</td><td>5</td></tr></table> <ul style="list-style-type: none"><li><b>two</b> correct points correctly plotted</li><li>line with gradient 2 but incorrect intercept</li><li>line with <math>y</math>-axis intercept 1 but incorrect gradient, not line <math>y = 1</math></li><li>a correct unruled line.</li></ul> <p>or</p> <p>Award 1 mark for either of these:</p> <ul style="list-style-type: none"><li>any <b>two</b> correct points found</li><li>correctly plotting <i>their</i> <b>three</b> points.</li></ul> | $x$      | -2 | 0 | 2 | $y$ | -3 | 1 | 5 | <p>Mark intention.</p> <p>For all marks, points may be implied by a line.</p> |
| $x$      | -2                                                                                                                                                                       | 0     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |    |   |   |     |    |   |   |                                                                               |
| $y$      | -3                                                                                                                                                                       | 1     | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |    |   |   |     |    |   |   |                                                                               |
| 20(a)    | $(5, 11)$                                                                                                                                                                | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |    |   |   |     |    |   |   |                                                                               |
| 20(b)    | $(7, 9)$                                                                                                                                                                 | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |    |   |   |     |    |   |   |                                                                               |
| 20(c)    | $(k =) 7$                                                                                                                                                                | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |    |   |   |     |    |   |   |                                                                               |

| Question | Answer                              | Marks    | Part Marks                                                                                                                                                                                                           | Guidance                                                                                                   |
|----------|-------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 21       | $0.6 - 0.2 \times (2 + 0.1) = 0.18$ | <b>1</b> |                                                                                                                                                                                                                      |                                                                                                            |
| 22       | 0.1 <b>or</b> $\frac{1}{10}$        | <b>1</b> |                                                                                                                                                                                                                      | Accept equivalent fractions or 10%.                                                                        |
| 23       | 1.005<br>400<br>–200 (kg)           | <b>2</b> | Award 1 mark for <b>one</b> or <b>two</b> correct sentences.                                                                                                                                                         | All <b>three</b> sentences correct for 2 marks.                                                            |
| 24       | $A = (4, 7)$                        | <b>2</b> | Award 1 mark for 4 correct <b>or</b> for 7 correct.                                                                                                                                                                  |                                                                                                            |
| 25       | $(x =) 0.5$ or $\frac{1}{2}$        | <b>3</b> | Award 1 mark for $10x + 2 = 2 - 20x + 15$ or better, with at most one term incorrect <b>and</b><br><br>Award 1 mark for correctly collecting <i>their</i> letters on one side and <i>their</i> numbers on the other. | Or better, e.g. $10x + 2 = 17 - 20x$<br><br>Implied by $10x + 20x = 2 - 2 + 15$ or better, e.g. $30x = 15$ |